

01

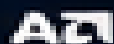
AI AGENT FOR MULTILINGUAL RURAL HEALTHCARE

VISION-LINK AI

Breaking Language
Barriers in Healthcare



Team InfinityTechian



AMD AI Hackathon

Building AI for Everyone, Everywhere.



02

THE PROBLEM



70%+ rural populations
lack access to quality
healthcare



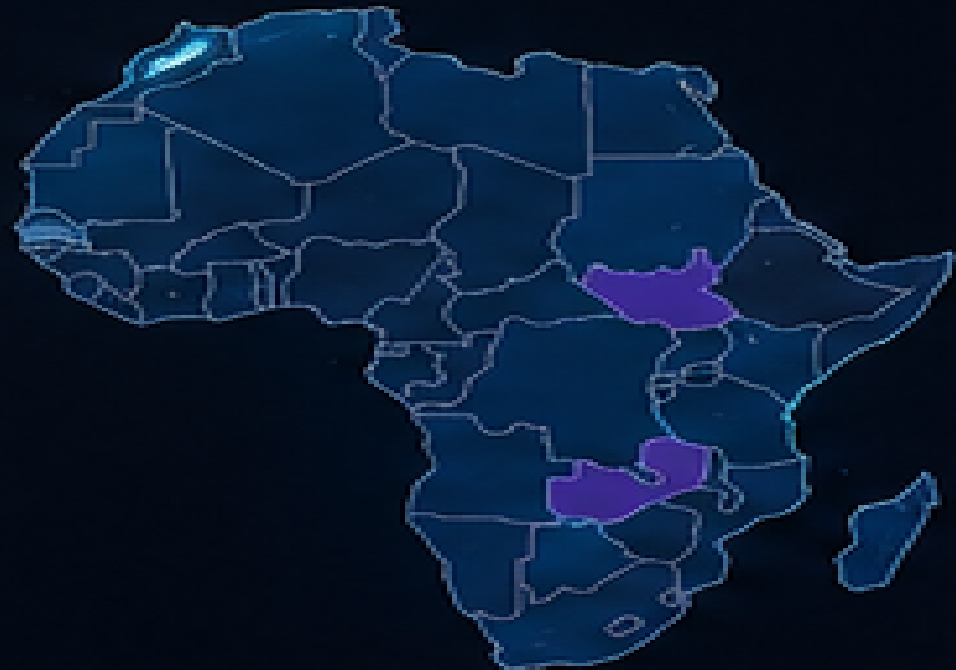
Language barriers
(Swahili, Hausa,
Luganda, etc.)



Limited internet →
Cloud AI not feasible



Shortage of trained
medical professionals



IMPACT



Delayed
Diagnosis



Miscommunication



Increased
Mortality

03

OUR SOLUTION

VISION-LINK AI AGENT

Offline Multilingual AI Assistant

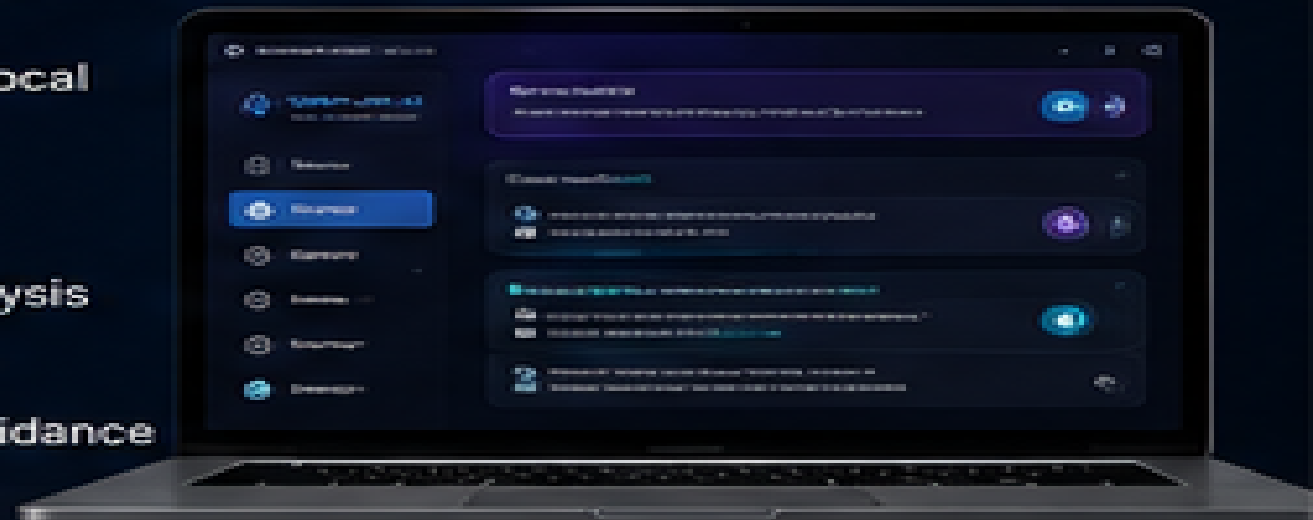


Understands local languages



Provides:

- Symptom analysis
- Health advice
- Emergency guidance

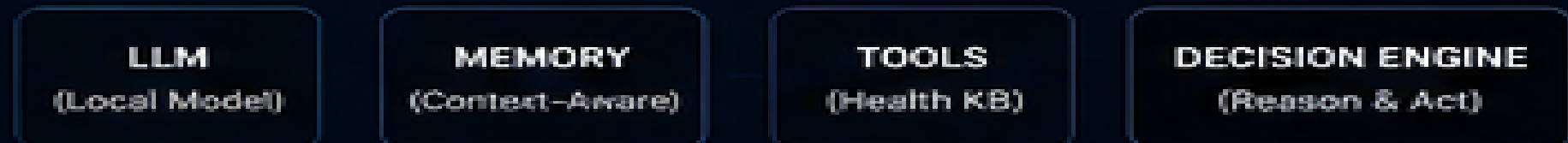


AI Doctor Assistant that works without internet

04

WHAT MAKES IT AN AI AGENT?

AGENTIC INTELLIGENCE

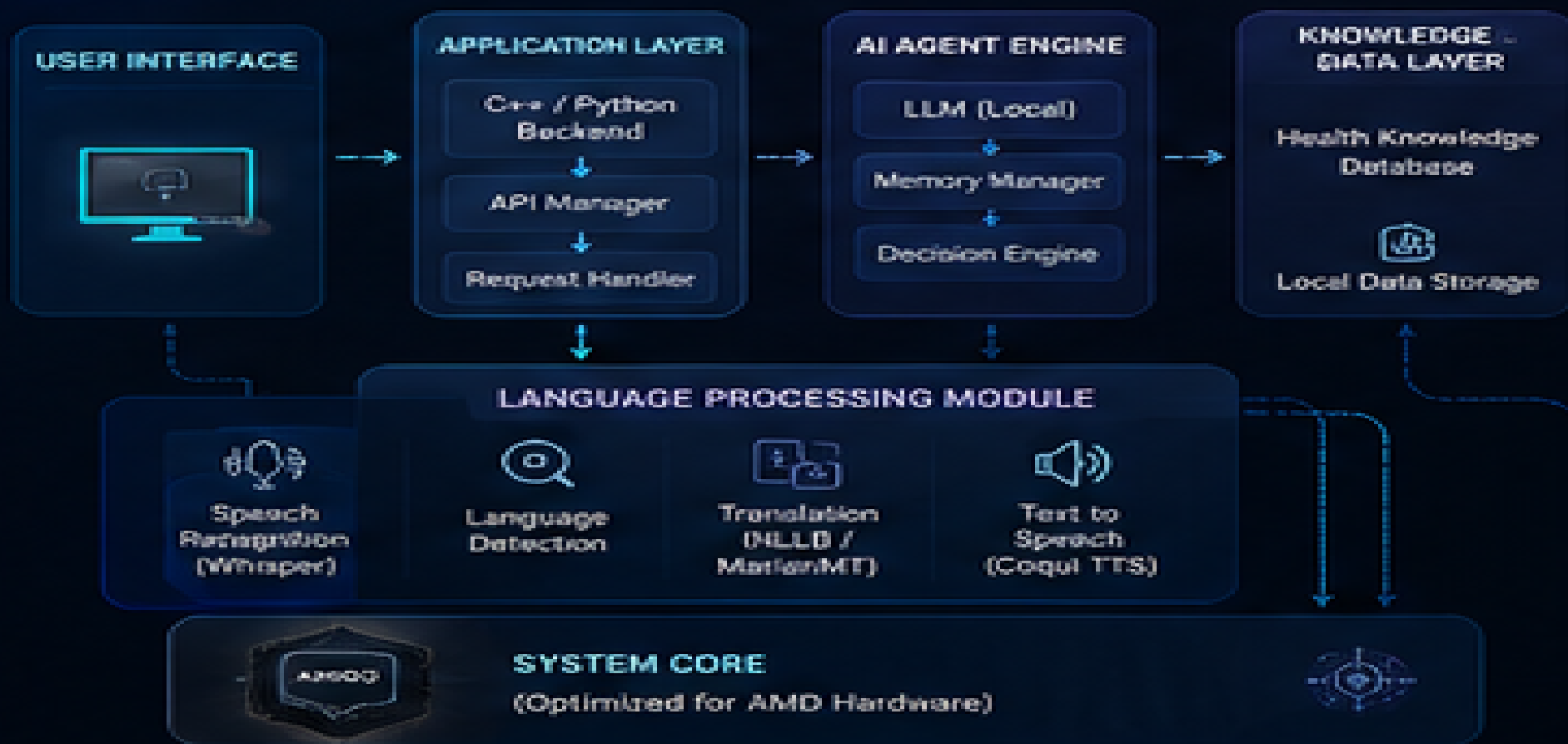


AGENT FLOW

User Query → Language Detection → Reasoning → Response

05

SYSTEM ARCHITECTURE



06

TECH STACK



FRONTEND

Qt (C++)



BACKEND

Python / C++



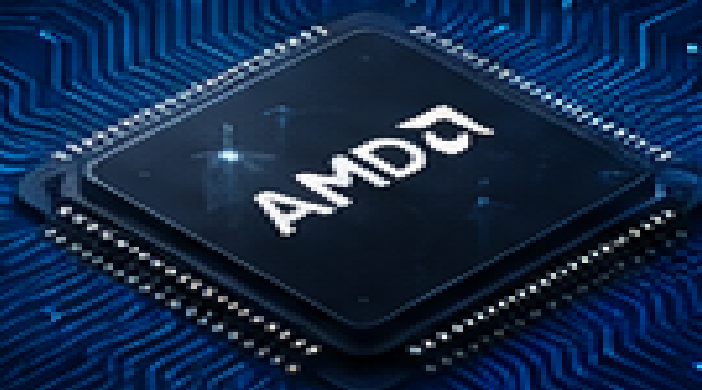
AI MODELS

- Quantized LLM (GGUF / ONNX)
- Whisper (Speech Recognition)
- MarianMT / NLLB (Translation)



DEPLOYMENT

- AMD Hardware Optimization
- Offline Execution
- Edge Ready



07

KEY FEATURES



Multilingual Support
(Swahili, Hausa, Luganda & More)



Offline Capability
(Works Without Internet)



Context Awareness
(Remembers & Understands)



Fast on Low Hardware
(Optimized for Edge Devices)



Health Knowledge Base
(Reliable Medical Information)



Voice Input / Output
(Speak Naturally)



08

INNOVATION / USP



Works without
internet



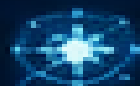
Focus on low-resource
African regions



Optimized for
edge devices



Combines
AI Agent + Healthcare
+ Multilingual



Not just AI — a deployable real-world solution



09

DEMO SCENARIO

1

User speaks in Swahili

2

AI translates → understands

3

Detects symptoms

4

Suggests possible conditions

5

Gives guidance



VISION-LINK AI

Your AI Health Assistant



"Ninaumwa kichwa na homa kwa siku mbili"



I have a headache and fever for two days.

Possible Conditions

- Malaria
- Typhoid Fever
- Viral infection

Guidance

Rest and drink plenty of fluids.
Take paracetamol for fever.
If symptoms worsen, visit a nearby health facility.



Tap to speak



10

REAL-WORLD IMPACT



Accessible healthcare
anywhere



Reduces dependency
on doctors



Helps NGOs &
rural clinics



Scalable globally



Better Health. Better Lives. Better Future.

11

FUTURE SCOPE



Add more
languages



Integrate IoT
health devices



Real-time doctor
connection
(when online)



Expand beyond Africa
→ Global



Continuously Evolving to Serve Humanity

12

THANK YOU!



VISION-LINK AI

AI FOR EVERYONE,
EVERYWHERE



Team InfinityTechian